

Exercise sheet 5: TranSMART solutions

Before starting the exercise: Install TranSMART with Docker, see GitHub.

TranSMART is available at <http://localhost:80>

user: admin, password: admin

1. Overview of the study

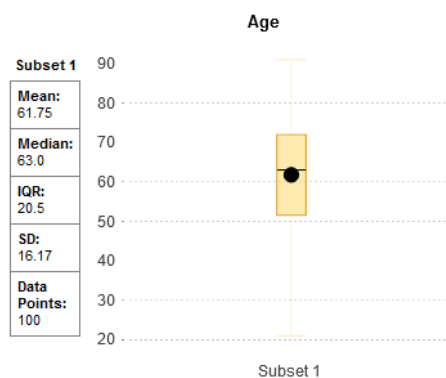
- a. Open the tree structure under "Public Studies" > "MIMIC" on the left under "Navigate Terms" and add the following variables to subset 1:

Age

The subset does not need to be saved.

- b. Go to the "Summary Statistics" tab. What is the mean and median age of the patients?

Solution: Mean: 61.75, median: 63



- c. What percentage of patients are male or female?

Solution: 43% female, 57% male

Category	Subset 1 (n)	Subset 1 (%n)
F	43	43.0 %
M	57	57.0 %

2. Comparing subsets

- a. Go back to the "Comparison" tab and delete the subset from task 1 by clicking on "Clear Panel."
- b. Now add new variables to the subsets. You can find the variables under "Public Studies" → "MIMIC" → "Clinical Data" → "Diagnosis."

Subset 1: Variable "E78.5" under "Diagnosis" → "Hyperlipidemia"

Subset 2: Variable "I48.0" under "Diagnosis" → "Atrial fibrillation"

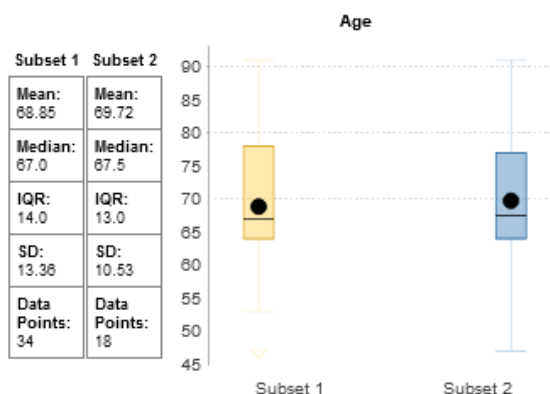
- c. Now go to the "Summary Statistics" tab:
How many patients are in each subset and how many patients have both diagnoses?

Solution:

Subject Totals		
Subset 1	Both	Subset 2
34	11	18

What is the mean age of patients suffering from hyperlipidemia or atrial fibrillation?

Solution:

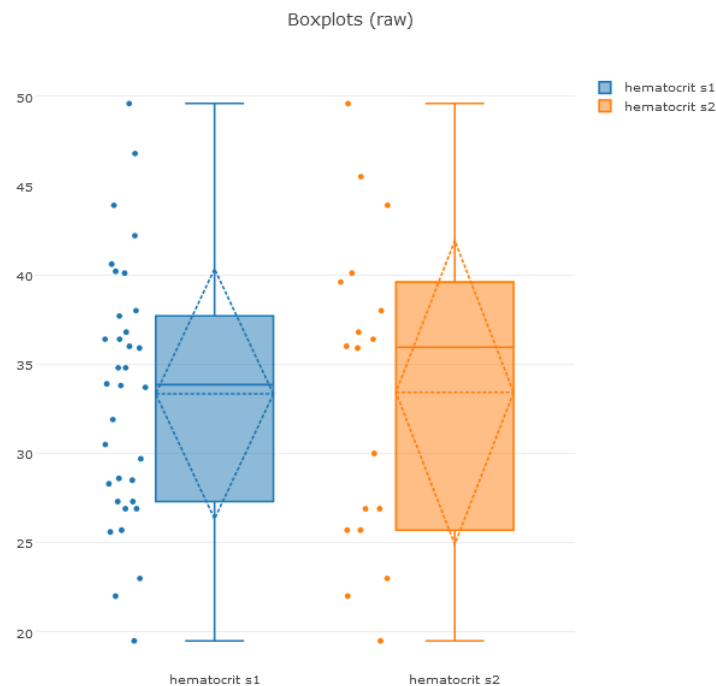


3. Box plot

Perform the task with the two subsets "hyperlipidemia" and "atrial fibrillation" from the previous task 2.

- Go to the "SmartR" tab and select "Boxplot Workflow" from the drop-down menu. Drag the variable "**hematocrit**" under "Clinical Data" → "Lab Values" into the left field "Numerical Variables." Click on "Fetch Data."
- After the data has been loaded, switch to "Run Analysis" and create boxplots with "*Raw Values*."

Solution:



- c. What is the mean hematocrit value for patients with hyperlipidemia or atrial fibrillation? For which diagnosis is the mean value higher?

Solution: Hyperlipidemia (subset 1): 33.85, atrial fibrillation (subset 2): 35.95

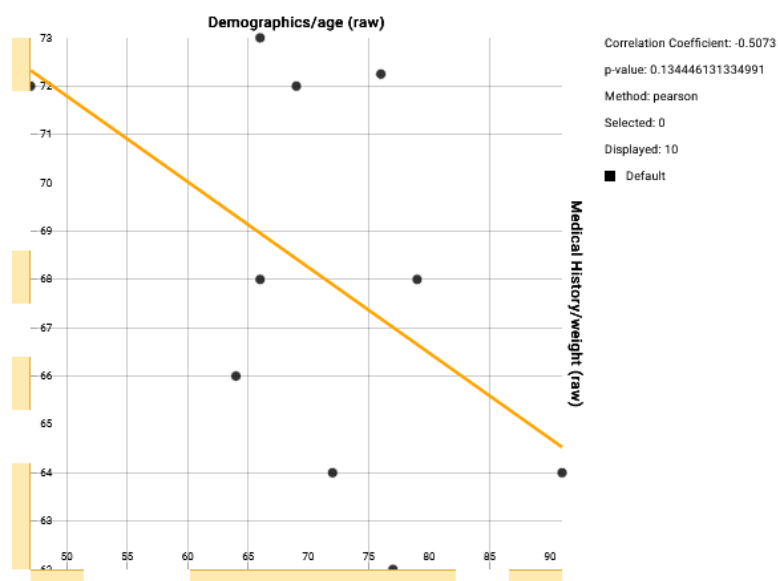
4. Correlation analysis

Perform the task with the "Atrial fibrillation" subset. The subset is changed in the "Comparison" tab.

Then go to the "SmartR" tab and select "Correlation Workflow."

- Select the numerical variables "age" and "weight." Click on "Fetch Data" and then go to the "Run Analysis" tab.
- Create a correlation analysis with "Raw Values" and the Pearson method.

Solution: (excerpt only)



- c. What is the correlation coefficient and what does it say about the age and weight of patients diagnosed with atrial fibrillation?

Solution: Coefficient=-0.5

=> The older the patients get, the less they weigh.

5. Survival time analysis

- a. First, perform this exercise with the "Hyperlipidemia" subset. The subset can be changed in the "Comparison" tab.
- b. Go to the "Advanced Workflow" tab and select "Analysis" from the drop-down menu.

Create a Kaplan-Meier curve with the following variables:

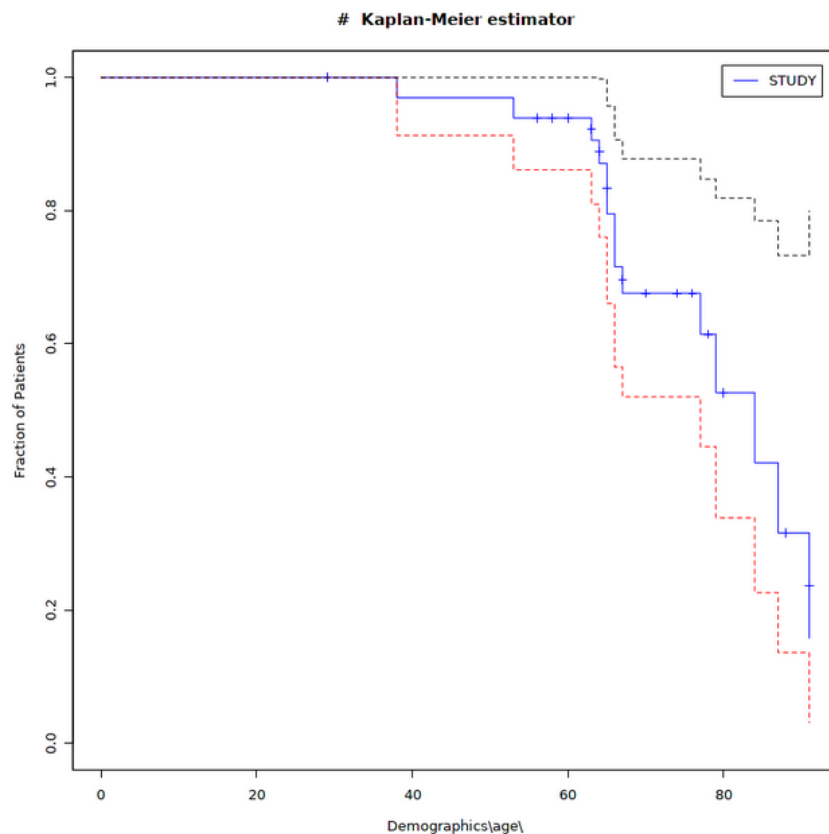
Time: age

Category: -

Censoring variable: alive

Solution:

Survival Curve

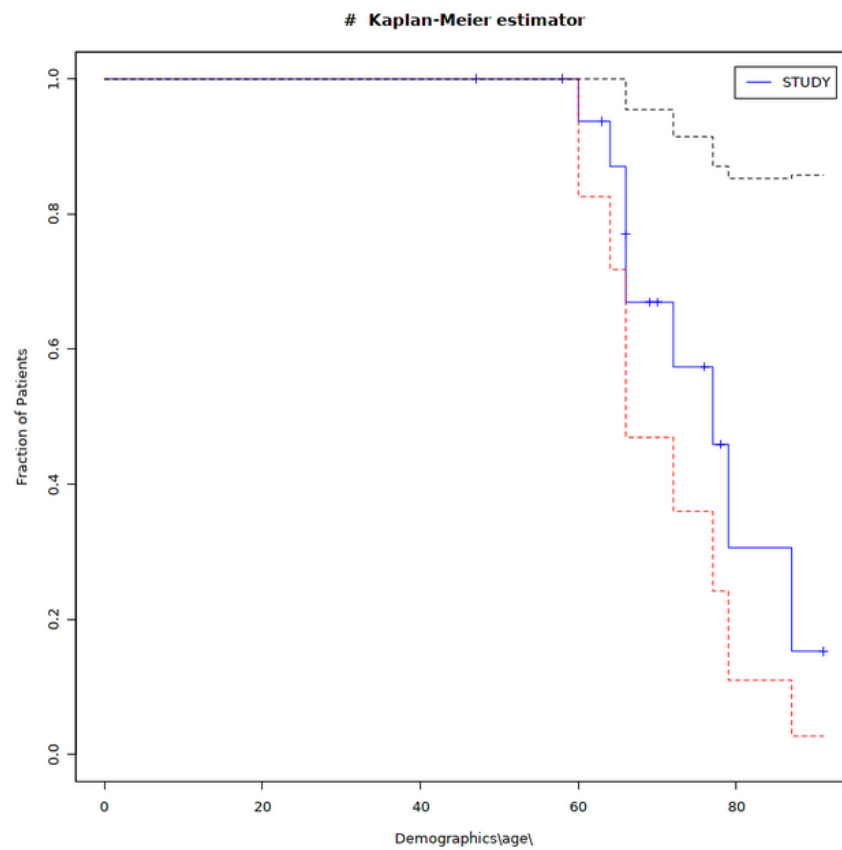


- c. Now change the subset to "Atrial fibrillation." The subset is changed in the "Comparison" tab.

Now perform a survival analysis again as in task 5.b.

Solution:

Survival Curve



- d. Now compare the two Kaplan-Meier curves for patients diagnosed with hyperlipidemia and atrial fibrillation. How many patients are still alive at age 80?

Solution: Hyperlipidemia = 50%, atrial fibrillation = 30%

